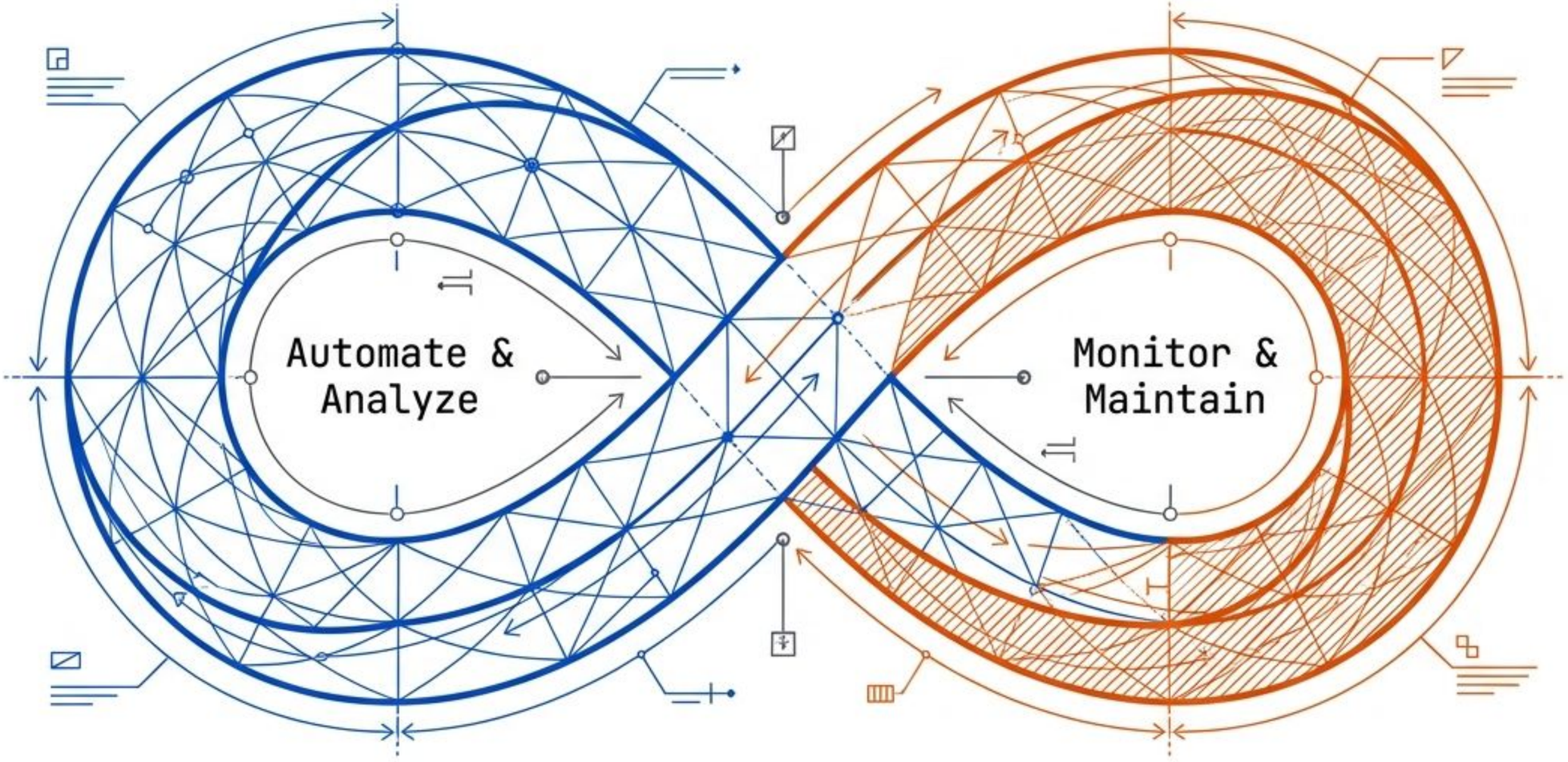
AWS Certified Data Engineer - Associate (DEA-C01)

Mastering Domain 3: Data Operations and Support



Focus: Operationalizing pipelines, data quality, and system health.

NotebookLM

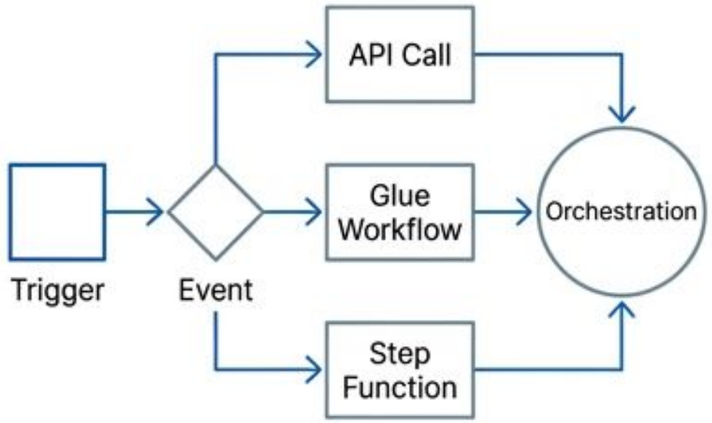
The Operational Pillar: Domain 3 Breakdown

22%

of Total Exam Score

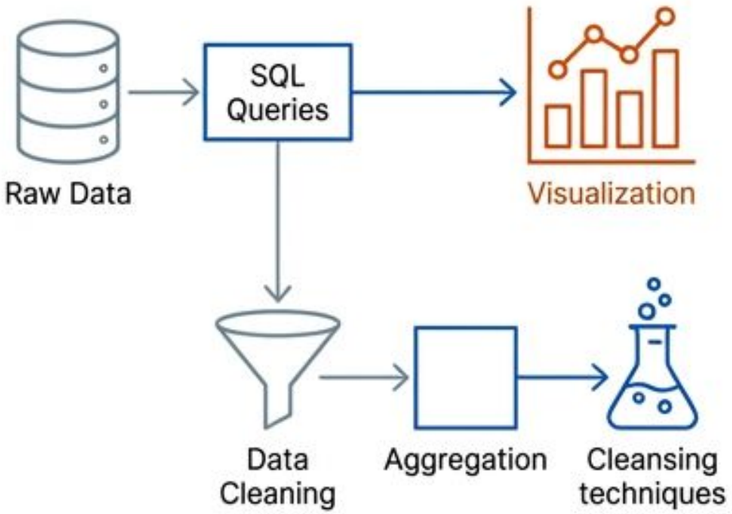
Task 3.1: Automate

Orchestration, API calls, Event-driven triggers, Glue Workflows, Step Functions.



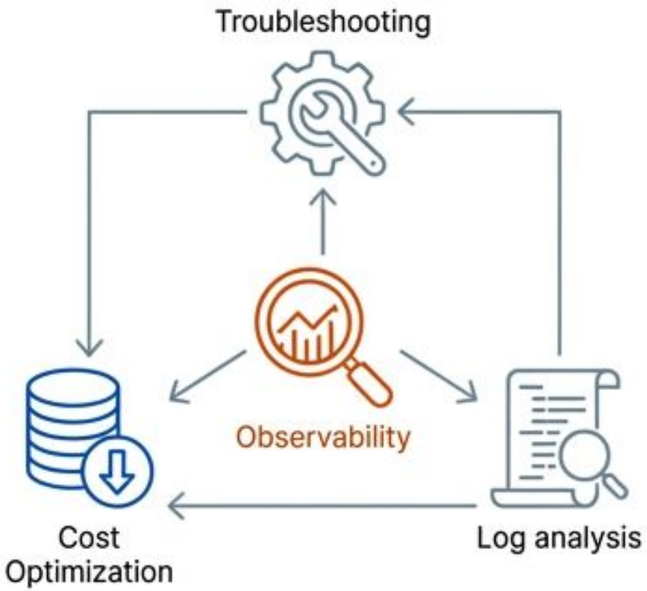
Task 3.2: Analyze

SQL queries, Visualization, Data Cleaning, Aggregation, Cleansing techniques.



Task 3.3: Maintain

Observability, Troubleshooting, Cost Optimization, Log analysis.



NotebookLM

Task 3.1: Orchestration Architecture

Choose Orchestration

Serverless & Visual

AWS Step Functions

- Visual Workflows
- State Management
- Key Feature: Map State (Parallel Processing)

Managed Open Source / Python

Amazon MWAA

- Complex Dependencies
- Portability (Airflow)

Glue Native

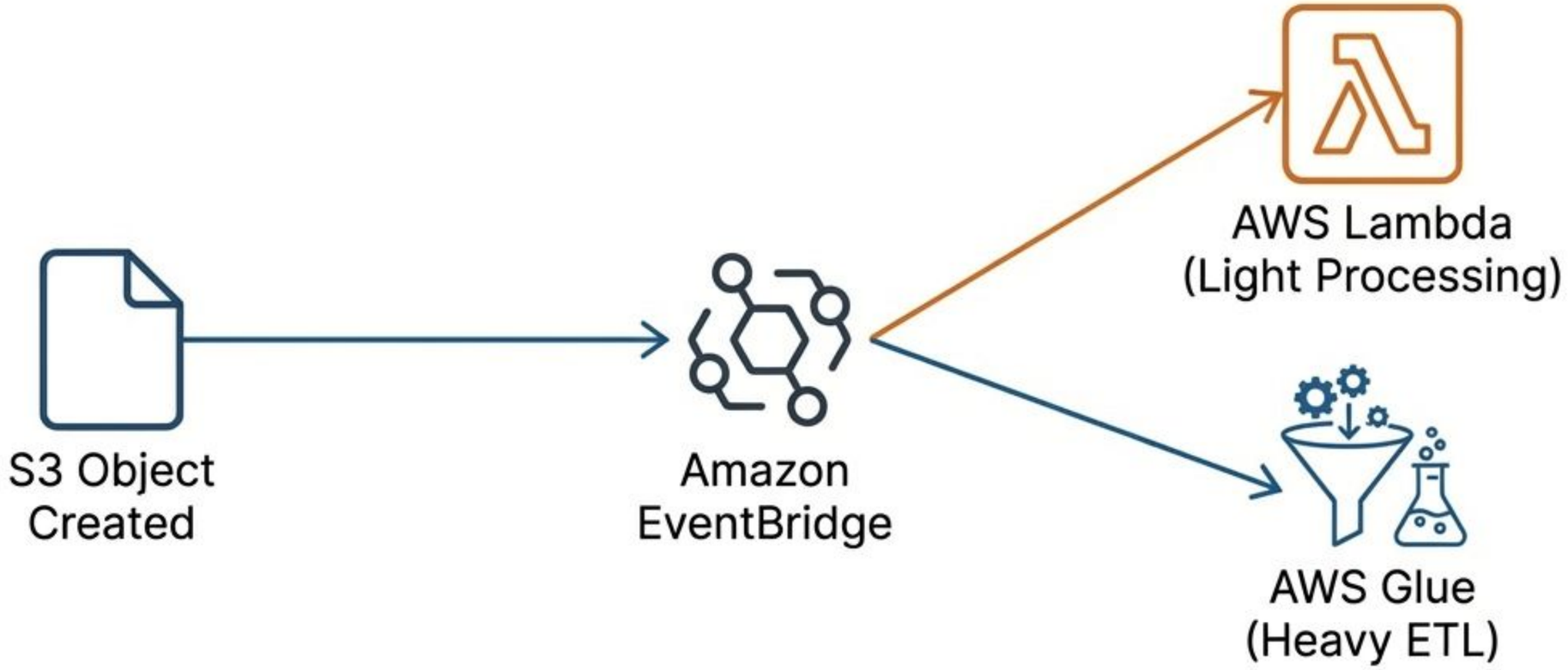
AWS Glue Workflows

- Tightly coupled Glue resources
- Triggers & Crawlers

Anti-Pattern:
Do not use Cron on EC2 for complex orchestration.

NotebookLM

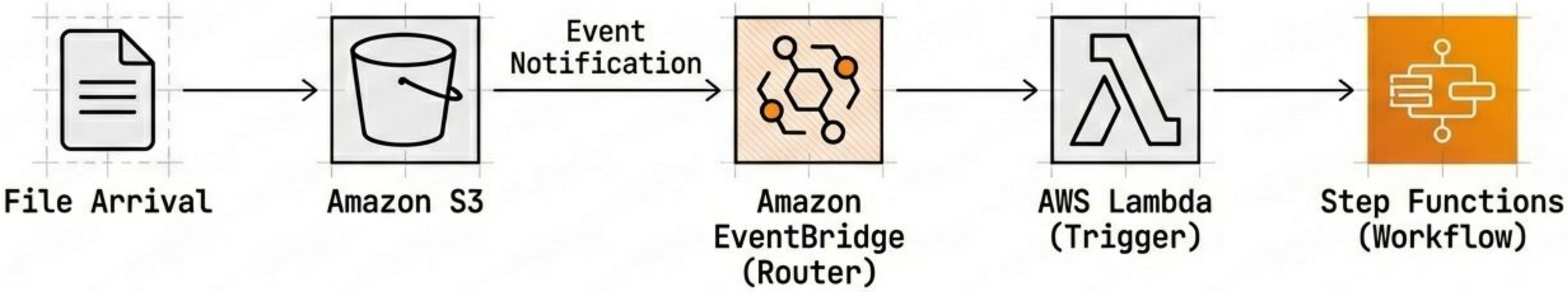
Event-Driven Architectures & Triggers



Design Principle: Replayability. Store raw events in S3 to allow reprocessing if downstream jobs fail.

NotebookLM

Event-Driven Automation



Reactive (Event-Based)	Batch / Migration
S3 Event Notifications	AWS DataSync
Immediate trigger on file upload	Scheduled synchronization
Good for: Individual file processing	Good for: Large-scale on-prem to S3 sync

NotebookLM

AWS Glue Operational Mechanics

Job Bookmarks
// Tracks processed data.
Enables incremental loading
to process only new deltas.

Triggers
// Activation methods:
Scheduled (Time-based),
Event (Job Succeeded/Failed),
or On-Demand.

Blueprints
// Parameterized templates.
Standardizes complex ETL patterns
across the organization.

NotebookLM

API Interaction & Scripting

Lightweight Logic

AWS Lambda

- Validation & API Calls
- Triggering Jobs
- Constraint: 15-min timeout
- ❌ ANTI-PATTERN: Heavy ETL / Large Data Cleaning

```
import boto3
client = boto3.client('s3')
response = client.list_buckets()
```

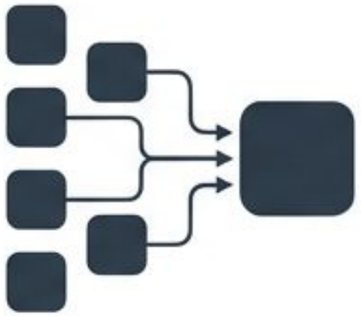
Heavy Lifting

AWS Glue & Scripting

- **Glue Shell Jobs:** Run Python scripts (Cheaper than ETL)
- **Glue ETL:** Distributed processing (Spark)
- **Boto3:** Python SDK for AWS API interaction

NotebookLM

Data Manipulation Techniques



Aggregation

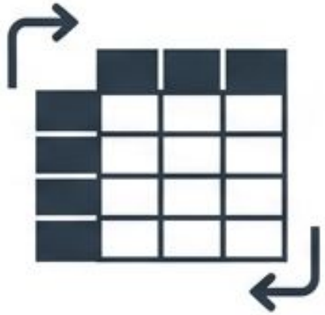
Sum, Avg, Count

Tools:
Redshift (GROUP BY),
Athena, OpenSearch

Rolling Average

Smoothing over a window

Tools:
Redshift (Window Functions),
QuickSight (Moving Calc)



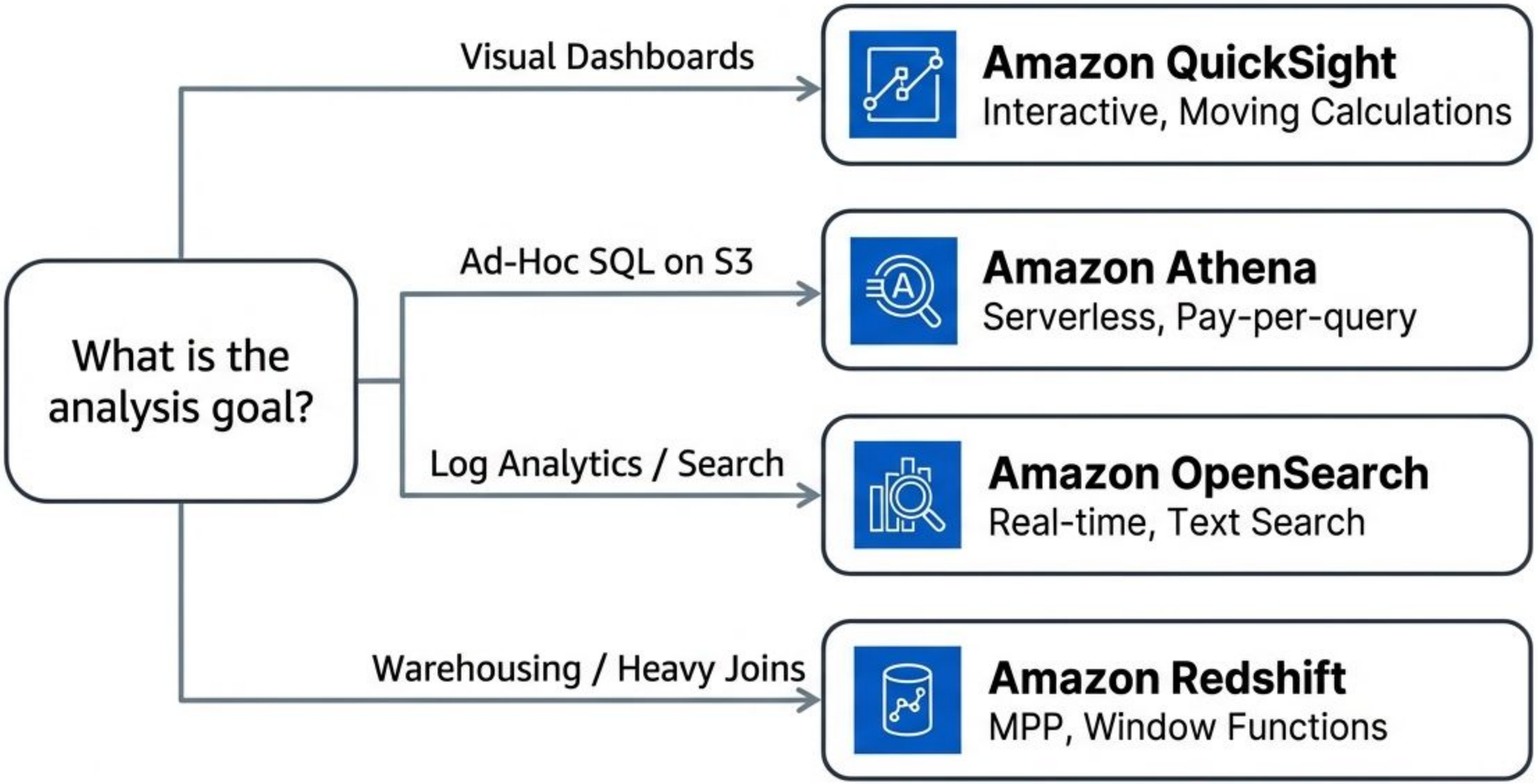
Pivoting

Rows to Columns

Tools:
Redshift (PIVOT),
QuickSight

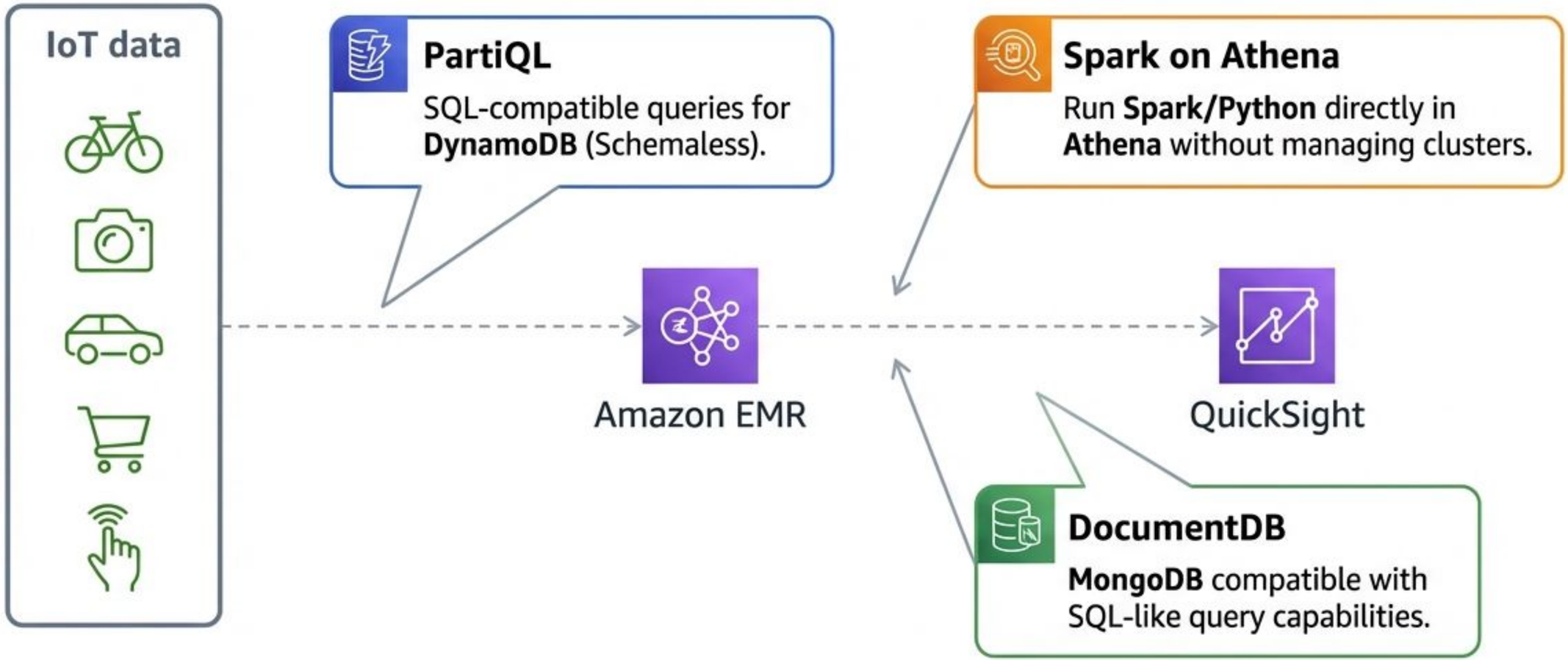
NotebookLM

Selecting the Right Analysis Tool



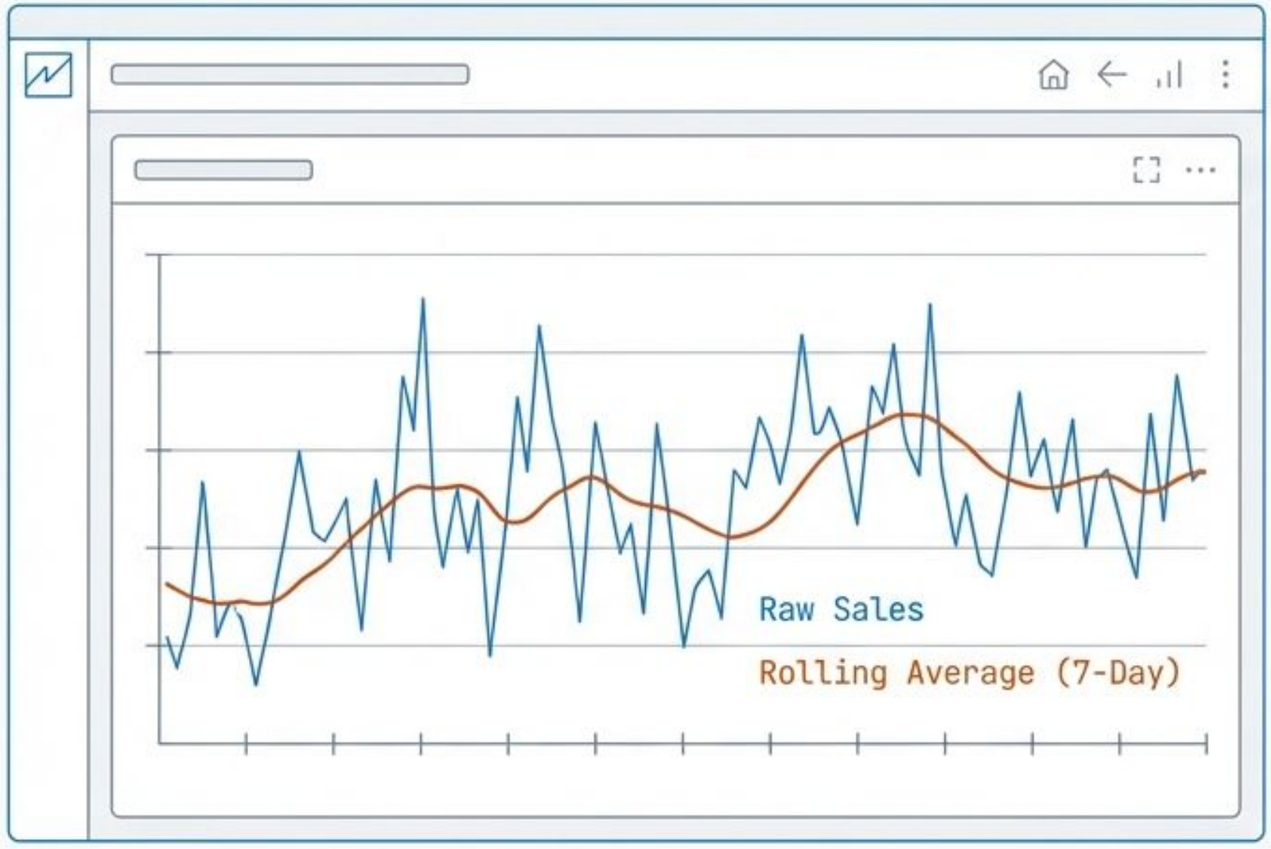
NotebookLM

Interoperability: SQL & Spark



NotebookLM

Visualization with Amazon QuickSight



**SPICE Engine*: Super-fast, Parallel, In-memory Calculation Engine.

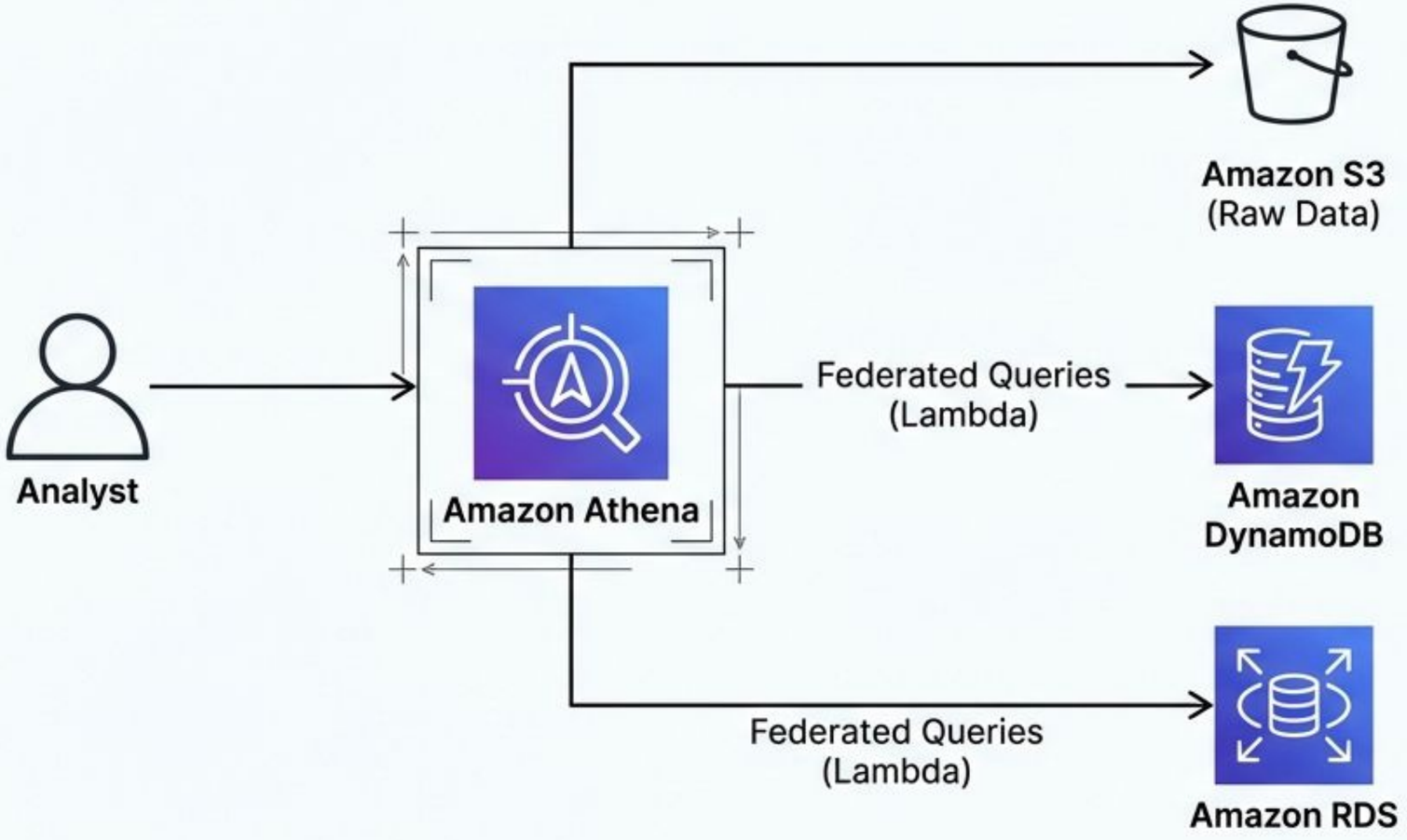
**Analytical Functions*: Built-in support for Rolling Averages, Anomaly Detection, and Forecasting.

**Integration*: Query S3 directly via Athena.

NotebookLM

Serverless SQL Analysis (Amazon Athena)

Inter Tight

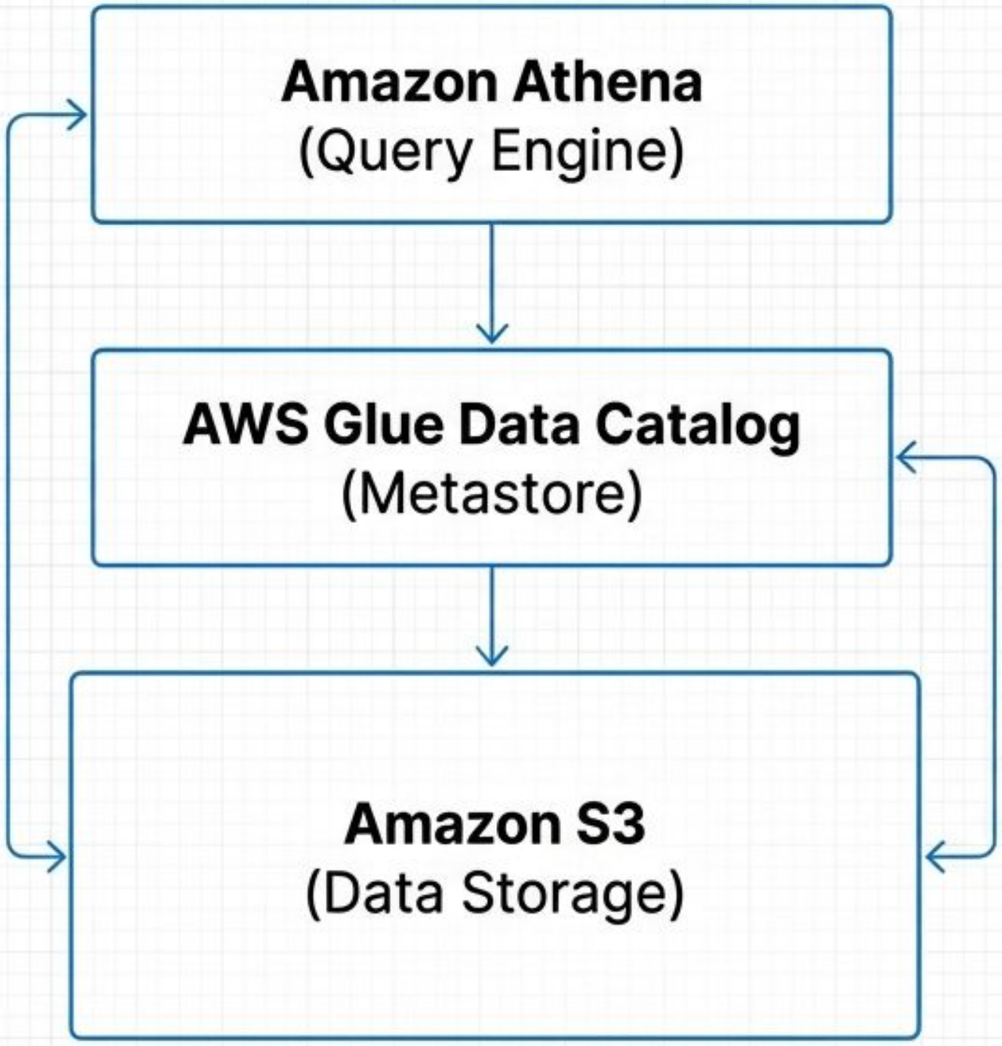


Key Capabilities

- Direct Analysis (No Load)
- CTAS (Create Table As Select)
- Partition Projection (Speed optimization)

NotebookLM

Analyzing Data: Athena & SQL Performance



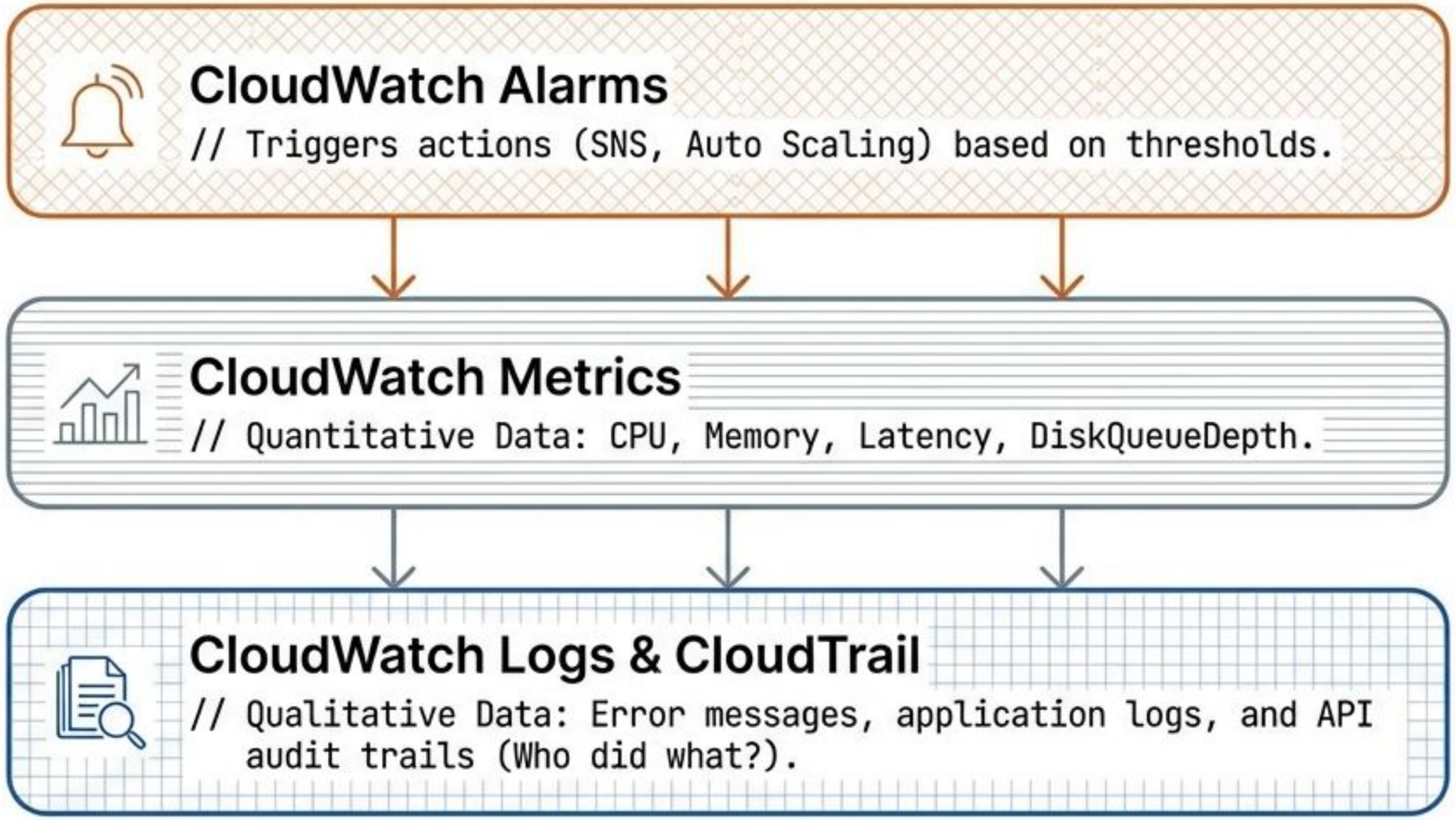
Performance Tuning Keys

1. **Partitioning:** Reduce scan size.
   └ Example: `year=2023 / month=01`
2. **Compression:** Use Splittable formats.
   └ Examples: `Snappy`, `Zlib`
3. **Format:** Columnar vs. Row.

   Parquet/ORC (Fast) ↔ **JSON/CSV (Slow)**
4. **CTAS:** Create Table As Select (Optimize downstream results).
5. **Federated Queries:** Connect to Redshift/RDS via Lambda.

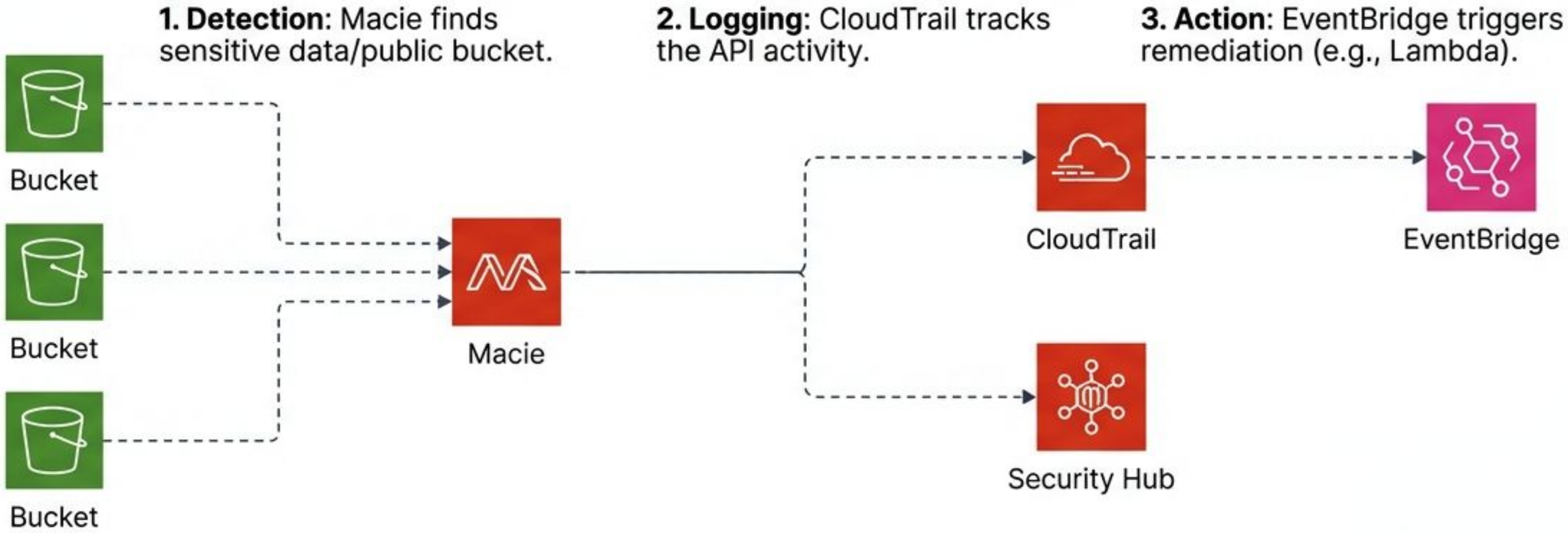
NotebookLM

The Observability Stack



NotebookLM

Security, Auditing, and Traceability

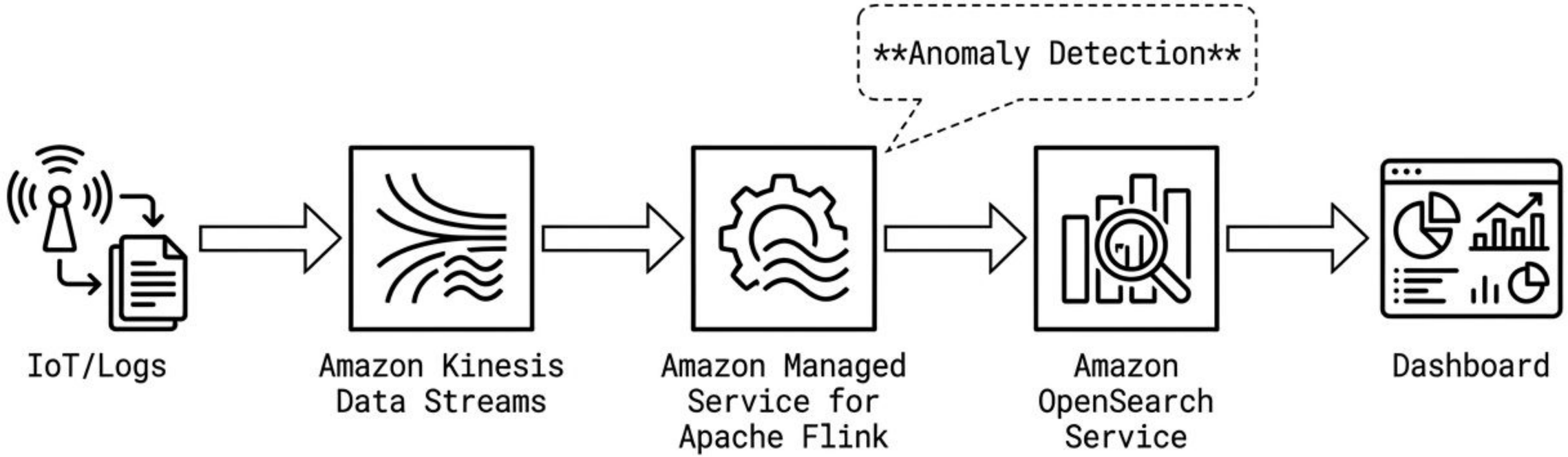


The Detectors

- **CloudTrail:** Who did what? (Inter Regular)
- **AWS Config:** Resource history. (in JetBrains Mono)
- **Security Hub:** Centralized alerts. (in JetBrains Mono)

NotebookLM

Real-Time Analytics & Anomaly Detection



Scenario Match
Scenario: Detect anomalies in real-time IoT data?
Answer: Amazon Managed Service for Apache Flink (formerly Kinesis Data Analytics).

NotebookLM

AWS Glue: Optimization & Troubleshooting

Error: OOM (Out of Memory)

Symptom: Job fails during heavy processing.

- Scale Out Workers (Switch G.1X -> G.2X)
- Optimize Code

Error: Data Skew

Symptom: Uneven distribution of keys causing load imbalance.

- Salt Keys
- Repartition Data

Issue: Slow Start-up Times

- Use Interactive Sessions
- Flexible Execution

Crawler Performance

- Use Incremental Crawls (Process only new folders).

NotebookLM

Tuning Redshift & Amazon MSK

Amazon Redshift

- **VACUUM:** Reclaims space from deleted rows and re-sorts data.
- **ANALYZE:** Updates statistical metadata for the query planner.

Amazon MSK (Kafka)

- **Replication Lag:** Monitor `ReplicatorBytesInPerSec` to ensure consumers keep up.
- **Disk Usage:** Scale storage when low. (Note: Express Brokers replicate 3x automatically).

NotebookLM

The Data Quality Defense

Gate 1
Pre-Processing

AWS Glue DataBrew

Visual Profiling, Distributions, Null checks.

Gate 2
During Processing

PyDeequ / Spark

Programmatic Validation jobs inside EMR/Glue.

Gate 3
Remediation

Lambda & Step Functions

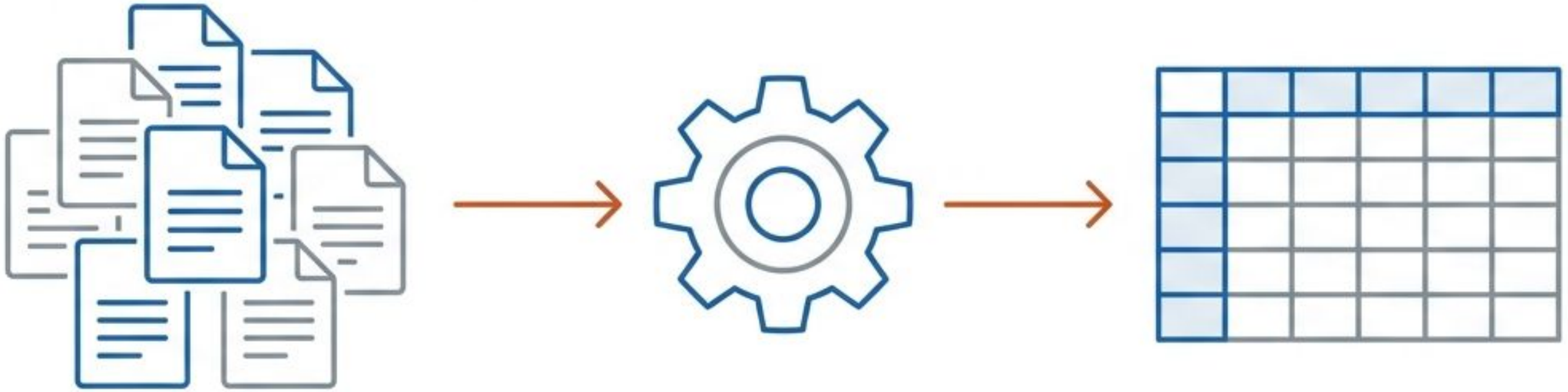
Route bad data to error queues; alert via SNS.

©2023, Amazon Web Services, Inc. or its affiliates. All rights reserved.

NotebookLM

Ensure Data Quality

Data Preparation & Quality Assurance



Raw Logs (JSON/CSV)
JetBrains Mono

AWS Glue DataBrew
JetBrains Mono

Cleaned Parquet Data
JetBrains Mono

- **Data Profiling:** Detect skew and distribution before processing.
- **Visual Cleaning:** No-code handling of nulls and PII masking.
- **Schema Evolution:** Use Glue Crawlers to detect new columns.

NotebookLM

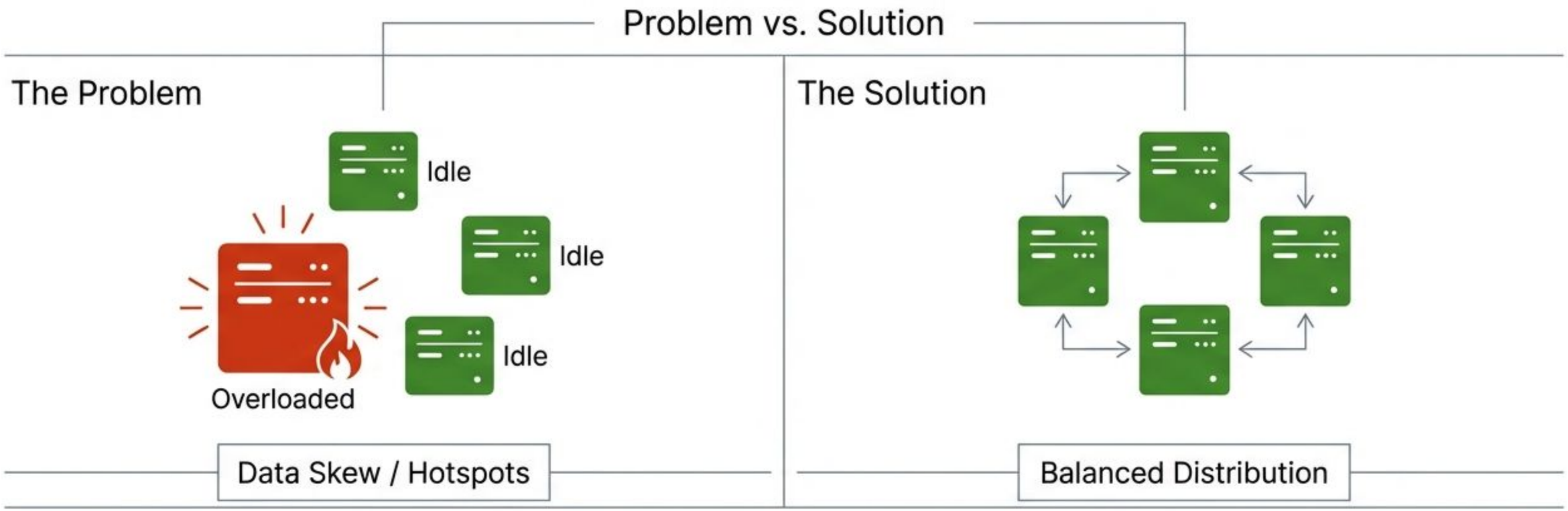
Ensure Data Quality

Inter Tight Trobleshooting Distributed Processing

Symptom	Diagnosis	Cure
One node working hard, others idle.	**Data Skew**	Salt keys, redistribute data, or choose high-cardinality partition keys.
Job fails with “Java Heap Space”.	**OOM (Out of Memory)**	Scale Out (more workers), Scale Up (larger worker type), or reduce shuffle partitions.
Slow Athena queries / High Listing overhead.	**Small Files Problem**	Compaction (Glue GroupFiles) to merge files into ~128MB chunks.

NotebookLM

Handling Data Skew



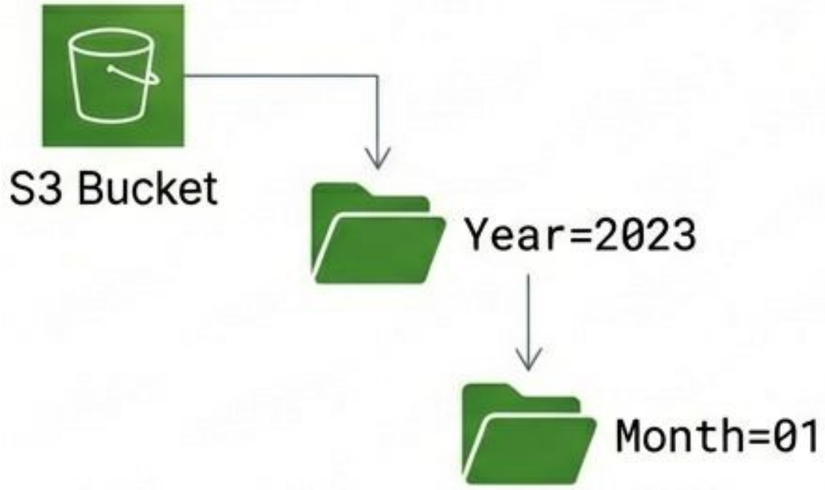
Remediation Mechanisms

- **Salted Keys**: Add random prefix to distribute keys.
- **Redshift Spectrum**: Use external tables for massive joins.
- **Adaptive Joins**: EMR/Redshift switch strategies (Hash vs. Sort Merge).

NotebookLM

Partitioning vs. Bucketing

Partitioning



Best for Pruning scans (`WHERE date = ...`).

Bucketing

Best for high-cardinality Joins/Aggregations.

Data Sampling Tools

- **S3 Select**: Query objects in place.
- **Redshift Spectrum**: External tables.
- **Stratified Sampling**: Spark technique.

NotebookLM

Top 5 Operational **Takeaways**

- **Prefer Managed Services:** Glue/DataSync > Custom Scripts
- **Think Event-Driven:** EventBridge/S3 Triggers > Cron Schedules
- **Decouple Compute & Storage:** Athena + S3 for flexible analysis
- **Automate Quality:** Embed Glue Data Quality checks in pipelines
- **Know Your Logs:** CloudTrail = Auditing (Who), CloudWatch = Performance (How)

NotebookLM